Zagazig University
Faculty of Computers & Informatics
Final Term Exam — Academic Year: 2024-2025

Department of Computer Science
Study Year/Level: First Year
Course Title: Discrete Structures
Course Code: CS103

Examination Date: 14/06/2025
Examination Starts: Morning
Allowed Examination Time: 3 Hours
Total Marks: 60 Marks

Student Name: **Student ID:**

- **Authorized Materials:**
- **Calculators allowed. No other additional materials or aides allowed.**
- **Instructions to Invigilators:**
- **The exam paper may be taken out of the examination room, but after half time.**
- **Instructions to Students:**
- **If you feel ill, let us know immediately.**
- **Write your answers in English.**
- **Write with a non-erasable pen, do not use red or green color.**
- **Show your working in detailed steps.**
- **You have 20 minutes to read the questions and, after that, 160 minutes to write your solutions.**
- **If you find a problem confusing or ambiguous, state your assumptions and solve.**
- **If you get stuck on a question, move on and come back later if you have time.**
- **Write your solutions directly into the exam booklet.**
- **You can obtain 60 points. You need 18 points to pass.**
- **Remember to time yourself.**

Enjoy! ****Questions 2 and 3 are optional (The student must choose and answer only one of them)****

Q1: Part I Fundamentals *Mandatory*

1. Write the following using set builder notation:
a) $A \subset B$
b) A^c

2. Write both an explicit and a recursive formula for each of the following sequences:
a) 1, -1, 1, -1, 1, …
b) 1, 1/3, 1/5, 1/7, 1/9, …

3. Suppose $X=\begin{bmatrix}0&0&1\\1&1&0\\1&0&0\end{bmatrix}$, and $Y=\begin{bmatrix}0&1&1\\1&1&0\\1&0&1\end{bmatrix}$, Find:
a) $Y \odot X$
b) $X \wedge Y$

4. Let A, B, and C be subsets of all integers, defined by: $A = \{ x \mid 0 < x < 5\}$
$B = \{3, 7, 19, 25\}$
$C = \{1,3, 5, 7, 11, 13, 17, 19, 23, 29\}$
a) What is $A \oplus B$?
b) Draw a Venn diagram of the sets A,B,C and label the elements in each set.

5. Determine whether the given set is the **power set** of some set. If the set is a power set, give the set of which it is a power set.
a) $\{\varnothing,\{a\},\{\varnothing, a\}\}$
b) $\{\varnothing,\{a\},\{\varnothing\},\{a,\varnothing\}\}$

6. Determine whether the set is **finite** or **infinite**. If the set is finite, find its **cardinality**.
a) $P(A)$, where A is the power set of $\{a,b,c\}$
b) $\{x \mid x \in \mathbf{N} \text{ and } 4x^2 - 8 = 0\}$

Q2: Part II Relations *Optional*

1. Let $A = \{1, 2, 3, 4\}$ and define a relation R on A where $aRb \Leftrightarrow |a - b| \leq 2$.
a) Draw the digraph of R.
b) Construct the matrix M_R
c) Find the in-degree and out-degree of each vertex.
d) Find the inverse (R^{-1}).
e) Find Dom(R) and Ran(R).
f) Determine the complement of R.
g) Is R an equivalence relation? [Justify].
h) Determine A/R.
i) Find R(4).
j) Find the transitive closure of R.

Q3: Part III Relations *Optional*

1. Let $A=\{1,2,3,4,5\}$, $R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (3,4), (4,3), (4,4), (5,5)\}$ and $S = \{(1,1), (2,2), (3,3), (4,4), (4,5), (5,4), (5,5)\}$.
a) Draw the digraph of R.
b) Is R reflexive/irreflexive [justify].
c) Is R transitive [justify].
d) Is S asymmetric/antisymmetric [justify].
e) Is S symmetric [justify].
f) Find $S \circ R$.
g) Find S^* if possible.
h) Find S^2
i) Is R a void relation?
j) Find R-S.

Q4: Part IV Graph *Mandatory*

1. Let G be a graph with vertices V = {1, 2, 3, 4, 5, 6}.

a) Draw the graph of the following adjacency matrix, where an index (i, j) in the matrix is 0 if there is no edge from vertex i to vertex j and 1 if there is.

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

b) Is the graph G connected? [justify]

c) Is the graph G simple? [justify]

d) Is the graph G complete? [justify]

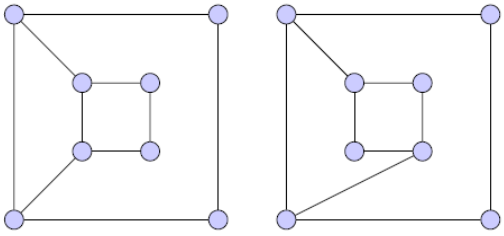
e) Is the graph G regular? [justify]

f) Does the graph G have an isolated vertex?

g) Does the graph G have a pendant vertex?

h) In graph G, is the sequence 2, 5, 6, 3, 4, 1, 2 is a Circuit walk? [justify]

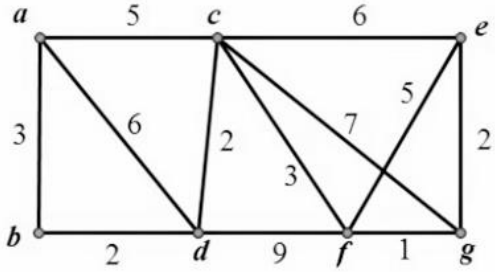
2. Are the following two graphs isomorphic? [justify]



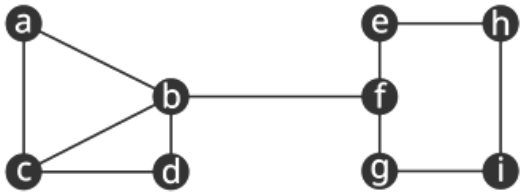
3. Use **Dijkstra algorithm** to find the following:

a) Shortest path from source vertex (a) to all other vertices. (***Please solve*** this question based on the instructions provided during the lecture**)**

b) Shortest path to visit **(e)** from **(a)** and the cost.



4. Apply **Fleury's algorithm** to find all possible Euler paths starting from the edge f → g.



Q5: Part V Trees *Mandatory*

1. Consider a complete 5-tree with 4 internal vertices.

a) What is the number of vertices?

b) What is the number of leaves?

2. Let A = {a, b, c, d, e, f} and T = {(a, b), (c, e), (f, a), (f, c), (f, d)}.

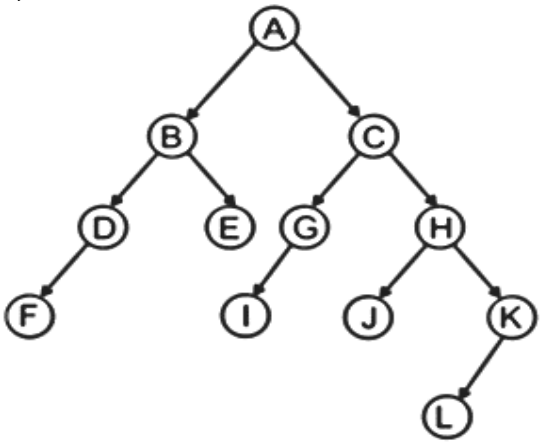
a) Is T a rooted tree? [justify your answer]

b) If T is a rooted tree, identify the root and draw the tree.

3. Consider the following tree (T) shown below. Answer the following questions:

a) Determine the **Inorder** traversal of the tree T.

b) Determine the **Postorder** traversal of the tree T.



4. Use binary tree to denote the following algebraic expression:

a) (5a - 3b)^2 * (3a + 5b)^3

b) (3 - (2 - (11 – (9 – 4)))) / (2 + (3 + (4 + 7)))

5. Convert the following algebraic expression into its equivalent **prefix** and **postfix** forms.

(x + (y – (x + y))) * ((3 / (2 * 7)) * 4)

6. Consider the following graph G, which has 13 edges. The weights of the edges are provided in the Table X below. Apply **Kruskal's algorithm** to find a Minimum Spanning Tree (MST) S of G.

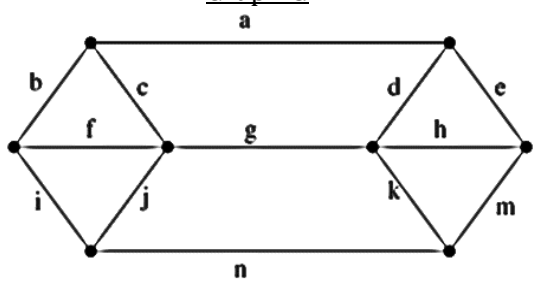
a) Draw the edges of S in the order in which they are added by Kruskal's algorithm. (If multiple valid solutions exist, list all possible MSTs).

b) Calculate the total cost of S.

Table X. Edge weights of graph G.

Edge	a	b	c	d	e	f	g	h	i	j	k	m	n
Weight	1	1	3	3	6	4	5	6	2	4	2	7	2

Graph G



Good Luck *Dr. Abduallah Gamal*